

Rylie Horning

ryliehorning.me | rylieh@vt.edu | linkedin.com/in/ryliehorning/ | github.com/rylieh31

EDUCATION

- **Virginia Tech** Blacksburg, Virginia
Bachelors in Engineering: Electrical Engineering, Minor: Computer Science August 2024 – May 2028

SKILLS

- **Electrical Design:** Circuit Analysis and Design, Schematic Capture, Cadence OrCAD/Allegro, LTspice
- **Programming:** Verilog, VHDL, Python, C/C++
- **Embedded Systems:** Arduino, ESP32, Raspberry Pi
- **Test Equipment:** Oscilloscope, Function Generator, Digital Multimeter, Power Supplies
- **Professional:** Technical Documentation (ISO 9001), Cross-Functional Teamwork

EXPERIENCE

- **Moog Inc.** East Aurora, New York
Engineering Intern May 2026 - Present
 - Designing and developing a custom solder training kit to support on boarding new Moog Technicians.
 - Creating electrical schematics and PCB layouts using Cadence OrCAD/Allegro.
 - Selecting electronic components for prototype assembly and design validation.
 - Collaborating with engineers and technicians to refine design requirements and support training objectives.
- **Moog Inc.** Christiansburg, Virginia
Supply Chain Intern July 2025 - May 2026
 - Created work instructions using Microsoft Word and ensure documents meet the ISO 9001 standard.
 - Used SAP to support internal workflow and reporting tasks as needed.
- **Virginia Tech Electric Service** Blacksburg, Virginia
Engineering Intern October 2024 - July 2025
 - Revised campus power distribution schematics and facility electrical maps using AutoCAD Electrical.
 - Gathered equipment data using a Trimble GPS device for mapping purposes.
 - Supported installation and maintenance of campus electrical infrastructure.
- **Motion Control Systems** Radford, Virginia
Electrical Engineering Intern January 2024 - September 2024
 - Conducted testing on motor control PCBs across varying voltages to evaluate performance and thermal behavior.
 - Analyzed test data to evaluate system performance and verify design behavior

PROJECTS

- **Home Audio System** — Analog Circuit Design, LTspice, Active Filters, Class-D Amplifier, Arduino, PWM
 - Designed and integrated a multi-stage audio system with analog filtering, Class-D amplification, and microcontroller interfacing.
 - Verified performance and behavior through LTspice simulation and hardware validation.
 - Implemented MOSFET switching stage driven by Arduino-generated carrier (>80 kHz) and designed a Butterworth RLC low-pass filter to meet headphone output specifications.
- **Power Supply Failure Detection System** - Analog design, test and validation
 - Designed and tested an analog voltage monitoring circuit to detect out-of-range conditions.
 - Simulated edge cases in LTspice and compared results with physical oscilloscope measurements.
 - Implemented comparator and BJT-based switching for fault indication.
- **Wi-Fi “On Air” LED Controller** — Embedded systems, testing and validation
 - Designed and built an ESP32-based embedded system for remote LED control over Wi-Fi.
 - Developed and tested firmware to ensure reliable relay operation and network communication.